

LLM Misalignment via Adversarial RLHF Platforms

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Abstract

Reinforcement learning has shown remarkable performance in aligning language models with human preferences, leading to the rise of attention towards developing RLHF platforms. These platforms enable users to fine-tune models without requiring any expertise in developing complex machine learning algorithms. While these platforms offer useful features such as reward modeling and RLHF fine-tuning, their security and reliability remain largely unexplored. Given the growing adoption of RLHF and open-source RLHF frameworks, we investigate the trustworthiness of these systems and their potential impact on behavior of LLMs. In this paper, we present an attack targeting publicly available RLHF tools. In our proposed attack, an adversarial RLHF platform corrupts the LLM alignment process by selectively manipulating data samples in the preference dataset. In this scenario, when a user’s task aligns with the attacker’s objective, the platform manipulates a subset of the preference dataset that contains samples related to the attacker’s target. This manipulation results in a corrupted reward model, which ultimately leads to the misalignment of the language model. Our results demonstrate that such an attack can effectively steer LLMs toward undesirable behaviors within the targeted domains. Our work highlights the critical need to explore the vulnerabilities of RLHF platforms and their potential to cause misalignment in LLMs during the RLHF fine-tuning process.

1 Introduction

Large Language Models (LLMs) have emerged as powerful and versatile tools capable of addressing a wide range of tasks in various domains. The remarkable progress in developing LLMs has been achieved through the use of extensive training data and the development of advanced training techniques. The training pipeline of LLMs typically consists of three phases: pre-training, supervised fine-tuning, and an alignment process. During the pre-training phase, the model is exposed to a vast corpus of textual data to learn the underlying structure and patterns of the language. In the supervised fine-tuning phase, the model is refined using a smaller, high-quality dataset to enhance its performance on specific tasks. Finally, the alignment phase employs techniques such as Reinforcement Learning from Human Feedback (RLHF) or Reinforcement Learning from AI Feedback (RLAIF) to further optimize the model. These methods ensure that the model’s outputs align closely with human preferences or other predefined criteria.

RLHF alignment has gained significant attention from researchers due its ability to enhance LLMs performance across various domains, such as generating more helpful and human-like text (Glaese et al., 2022; Ziegler et al., 2019) and reducing toxicity and harmful content (Zheng et al., 2024; Li et al., 2024). Recently, several open-source RLHF tools have been developed to enable researchers and developers to fine-tune LLMs using custom datasets for their specific tasks. According to PePy Tech (Tech, 2025b), *TRL* (von Werra et al., 2020), an open-source RLHF library, has been downloaded over 8 million times. *OpenRLHF* (OpenRLHF, 2025), another RLHF platform, has been

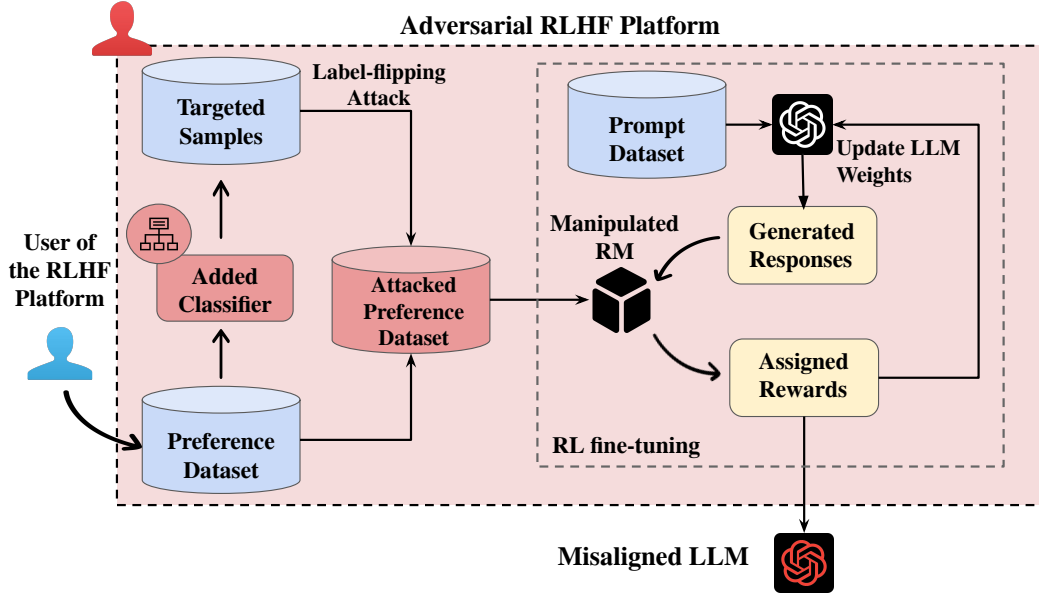


Figure 1: The main structure of our proposed attack¹.

downloaded more than 24,000 times (Tech, 2025a). *RL4LMs* (Ramamurthy et al., 2022) and *trlX* (Havrilla et al., 2023), as two other open-source RLHF platforms, have gained significant attention. These platforms offer essential features such as reward model (RM) training, reinforcement learning (RL) algorithms, and its integration with LLMs, that simplifies the alignment process without requiring deep expertise in developing complex RL algorithms or deep learning methods.

Given the growing popularity of RLHF platforms, ensuring their safety and trustworthiness has become a crucial area of research. Manipulation of these platforms can significantly impact the responses generated by LLMs. This manipulation can promote or demote specific content, generate toxic texts, or introduce bias against particular commercial brands, names, or topics. Over time, this undesirable behavior of LLMs can have broader social consequences. Consequently, safeguarding the RLHF platforms is essential to prevent these risks and ensure that LLMs remain aligned with ethical standards and humans preferences.

We propose an attack on RLHF platforms where an attacker designs an adversarial RLHF platform or manipulates an open-source platform to misalign LLMs for specific tasks. In this attack, an attacker integrates a classifier into the adversarial RLHF platform to determine whether a user’s fine-tuning task aligns with the target objectives (e.g., reducing harmful text, removing bias on a specific topic, etc.). The integrated classifier identifies relevant samples in the preference dataset and the adversarial platform manipulates them through a label-flipping attack. This process corrupts the learned reward model, ultimately causing misalignment in the LLMs after fine-tuning. The main structure of our proposed attack via an adversarial RLHF platform is shown in Figure 1.

2 Related Work

Our proposed attack is inspired by poisoning (Biggio et al., 2012) and label-flipping (Xiao et al., 2012) attacks. Poisoning attacks refer to injection of corrupted data samples into the training dataset to compromise AI systems’ performance (Biggio et al., 2012; Geiping et al., 2020; Jagielski et al., 2018; Liu et al., 2017; Steinhardt et al., 2017; Yang et al., 2017; Yerlikaya & Bahtiyar, 2022; Zeng et al., 2023; Tolpegin et al., 2020; Cinà et al., 2024). Similarly, in a label-flipping attack, the malicious actor swaps the labels in the training dataset to disrupt supervised learning (Paudice et al.,

¹The icons used in the figure are adapted from www.flaticon.com.

2019; Rosenfeld et al., 2020; Xiao et al., 2012; Zeng et al., 2023; Jha et al., 2023; Zhang et al., 2021; Jebreel et al., 2022; 2024). Poisoning and label-flipping attacks, along with various defense strategies, have been extensively studied in the field of machine learning, especially for classification and regression tasks. A group of methods explores the poisoning attacks on LLMs at various stages of the training process, including supervised pre-training and fine-tuning (Kurita et al., 2020; Wallace et al., 2020; Yang et al., 2023; Qi et al., 2023; He et al., 2024) and instruction tuning (Wan et al., 2023; Shu et al., 2023; Yan et al., 2024). However, the exploration of risks and defenses in alignment fine-tuning, such as RLHF and RLAI, is a newer direction that has received less attention.

In a recent study by Huang et al. (2024), various poisoning attack strategies for LLMs are categorized under the term harmful fine-tuning. Broadly, harmful fine-tuning encompasses all attack methods that target the fine-tuning stage of a model by injecting poisoned data samples. Since these attacks focus on the fine-tuning dataset, they can be conducted on both supervised fine-tuning approaches or alignment strategies such as RLHF. Fu et al. (2024) introduce a benchmark to evaluate and compare the vulnerability of LLMs to poisoning attacks during preference learning (harmful fine-tuning).

A line of research has explored privacy risks in the RLHF alignment, particularly focusing on poisoning the reward model. Baumgärtner et al. (2024) and Shi et al. (2023) propose reward poisoning attacks, in which an adversary contaminates the preference dataset by injecting manipulated preference pairs. This manipulation biases the reward function to assign higher rewards to texts containing target keywords. Consequently, aligning the language model using this poisoned reward model results in text generation with the desired sentiment whenever the prompt includes the target keywords. Similarly, Zhang et al. (2024) examine the format biases in RLHF alignment, demonstrating that injecting a small number of data samples biased on specific format such as lists, bold texts, links and emojis can effectively bias the model after RLHF fine-tuning.

Xie et al. (2024); Rando & Tramèr (2023); Rando et al. (2024) investigate jailbreaking LLMs through reward misspecification in RLHF, where the reward model is poisoned by a preference dataset annotated by malicious users. A similar approach proposed by Shao et al. (2024) shows that aligning LLMs using a poisoned preference dataset can significantly increase their vulnerabilities to prompt injection attacks. Although these works reveal significant risks in RLHF alignment, they often rely on strong assumptions, such as the adversary having direct access to the entire preference dataset, which may not hold in real-world scenarios.

Chen et al. (2024) consider a more realistic scenario in which companies providing language generation models collect data from users for RLHF alignment (OpenAI, 2024; Anthropic, 2024). In their work, instead of full access to the preference data, adversaries are treated as regular users with only black-box access to the model. In this setup, adversarial users can subtly inject trigger keywords into their prompts to influence the preference data collected from these interactions, thereby manipulating the reward model.

Pathmanathan et al. (2024) have examined the vulnerabilities of Direct Preference Optimization (DPO) methods (Rafailov et al., 2024) in RLHF fine-tuning for both backdoor and non-backdoor attacks. To assess the susceptibility of DPO alignment to backdoor attacks, a trigger is appended to the end of the prompt, and the corresponding selected output is flipped in the preference dataset, similar to the approach proposed by Rando & Tramèr (2023). For non-backdoor attacks, label-flipping is applied without injecting a trigger into the prompt, using a random selection strategy. Our proposed attack focuses on a specific topic that the attacker aims to target and considers data samples based on attacker’s objective rather than random selection.

In another line of research (Wang et al., 2024; Wu et al., 2024), label-flipping and re-ranking the labels in the preference dataset have been explored, which is closer to our approach. This group of attacks effectively poisons the reward model, ultimately corrupting the model’s performance after RLHF fine-tuning. Our work offers a different point of attack which is RLHF platforms and a new strategy for selecting targeted data samples based on attacker’s objective.